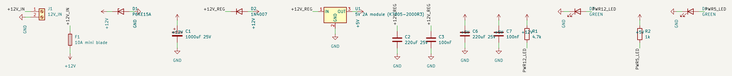
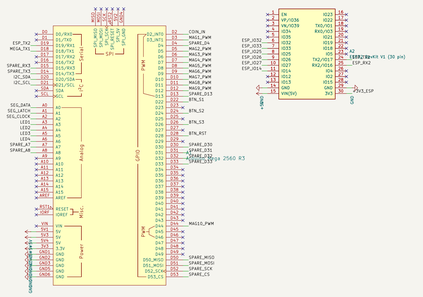
POWER: 12V INPUT, FUSE, SURGE PROTECTION, 12V -> 5V REGULATOR



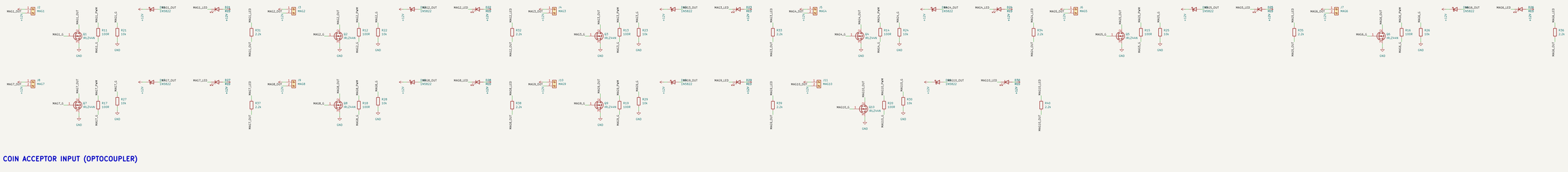
CONTROLLERS: ARDUINO MEGA 2560 (shield) + ESP32 DEVKIT V1



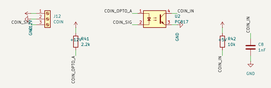
SERIAL: MEGA TX1 (5V) -> DIVIDER -> ESP32 RX2 (3.3V)



10x ELECTROMAGNET DRIVERS (IRLZ44N + FLYBACK DIODE + ON LED)



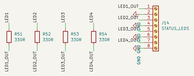
COIN ACCEPTOR INPUT (OPTOCOUPLER)



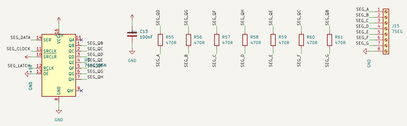
BUTTONS: SPEED1 / SPEED2 / SPEED3 / RESET



STATUS LEDS



7-SEGMENT DISPLAY DRIVER (74HC595)



EXPANSION HEADER



POWER FLAGS



Generated from gen/netlist.py — firmware pin map unchanged
Arduino Mega 2560 R3 shield + ESP32 DevKit V1 (30 pin)

Sheet: /
File: reflex-machine.kicad_sch

Title: Reflex Machine Controller Shield

Size: User	Date: 2026-09-11	Rev: 1.0
KiCad 8.0.6		Id: 1/1